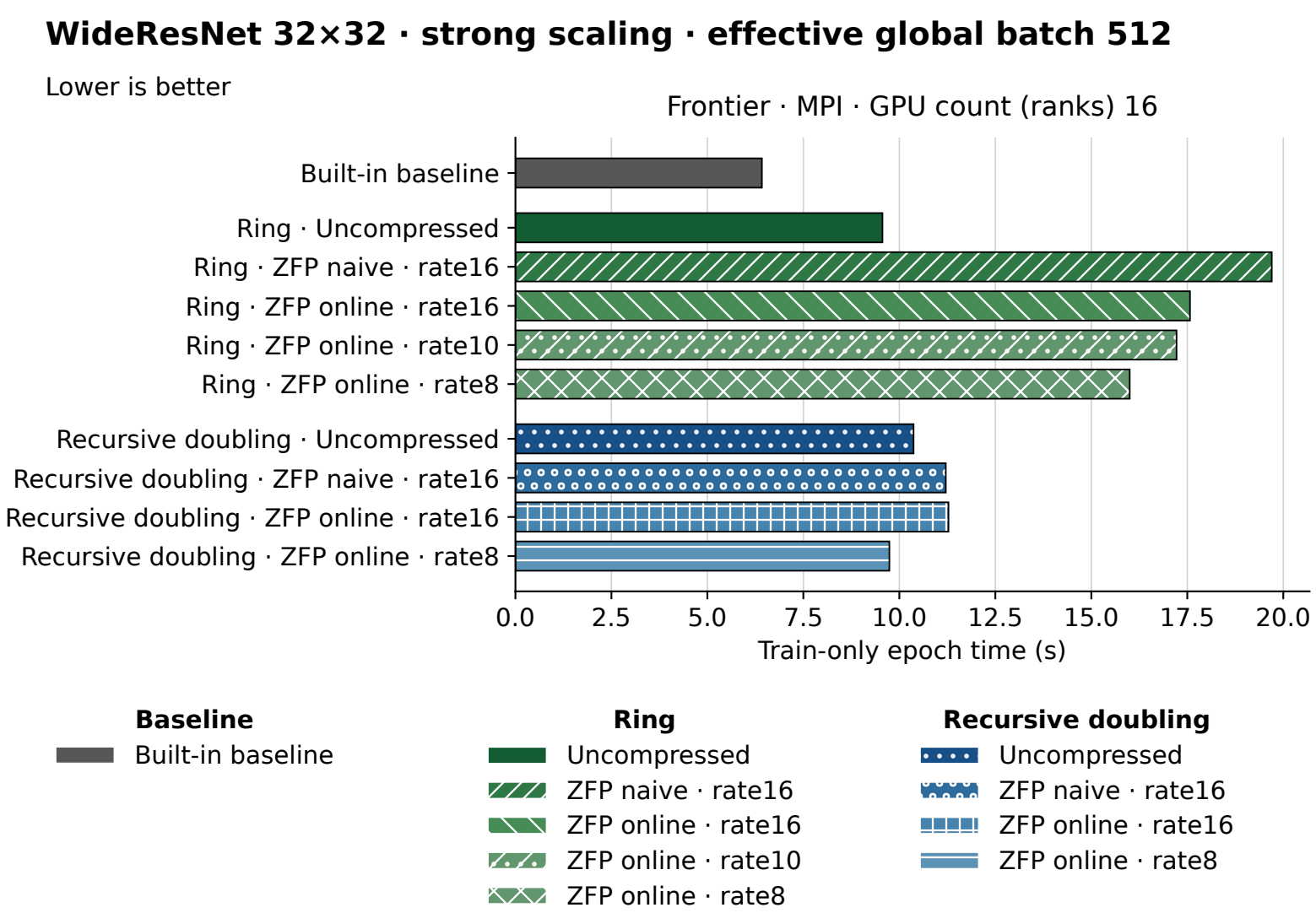

WideResNet 32×32 · strong scaling · effective global batch 512
Lower is better
Frontier · MPI · GPU count (ranks) 16
Built-in baseline
Ring · Uncompressed
Ring · ZFP naive · rate16
Ring · ZFP online · rate16
Ring · ZFP online · rate10
Ring · ZFP online · rate8
Recursive doubling · Uncompressed
Recursive doubling · ZFP naive · rate16
Recursive doubling · ZFP online · rate16
Recursive doubling · ZFP online · rate8
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Train-only epoch time (s)
Baseline
Built-in baseline
Ring
Uncompressed
ZFP naive · rate16
ZFP online · rate16
ZFP online · rate10
ZFP online · rate8
Recursive doubling
Uncompressed
ZFP naive · rate16
ZFP online · rate16
ZFP online · rate8
Epoch 1 excluded · missing metrics are not zero · r = ZFP rate